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Lei Chen Ph.D.

PROFILE

Lei Chen is a Postdoctoral Researcher at **the Zhejiang University-University of Edinburgh Institute, International Campus, Zhejiang University**. He received his **Ph.D. in Biochemistry and Molecular Biology (Biomaterials)** from **East China Normal University**. His current research focuses on biomaterials, wound healing and stem cell-based therapy for tissue regeneration.

EDUCATION

- **East China Normal University** *Sep. 2019 – Jun. 2025*
  Ph.D. in Biochemistry and Molecular Biology (Biomaterials) Shanghai, China
  - GPA: 3.60/4.00
  - Supervisor: **Prof. Qiang Zhang**
- **Zhejiang A&F University** *Sep. 2015 – Jun. 2019*
  B.Sc. in Applied Chemistry Hangzhou, China
  - GPA: 88.6/100 (Ranked 1st in major)
  - Awarded provincial and university-level scholarships for academic excellence

PROJECTS

- **Photosynthetic Microalgae–MSC Symbiotic Hydrogel for Chronic Non-Healing Wounds** *2025 – Present*
  Postdoctoral Project, Zhejiang University-University of Edinburgh Institute
  - Investigating a **photosynthetic microalgae–MSC symbiotic system** to actively remodel the pathological microenvironment of chronic non-healing wounds.
  - Developing an **in situ-gelling biohybrid hydrogel** via mild enzymatic crosslinking to co-encapsulate microalgae and mesenchymal stem cells for sustained light-triggered oxygen generation.
  - Studying how photosynthetic oxygenation regulates **MSC energy metabolism, HIF-1α signaling, and extracellular vesicle secretion** under persistent hypoxia.
  - Elucidating how **photosynthetic potentiation and dual-source bioactive component synergy** promote angiogenesis, inflammation resolution, and tissue reconstruction.
  - Evaluating therapeutic efficacy in multiple **preclinical wound models** to establish a new strategy for active modulation of the hypoxic wound microenvironment.
- **Superhydrophobic Nanocellulose Aerogel for Hemostatic Applications** *2020 – 2023*
  Material Synthesis & Characterization
  - Developed a novel **bio-aerogel** from bacterial cellulose nanofibers with superhydrophobic properties.
  - Achieved **rapid clot formation** in *in vitro* and *in vivo* tests with minimal side effects.
  - Utilized **SEM, TEM, and DMA** to characterize morphological and mechanical properties.
- **Inorganic-Organic Composite for Advanced Hemostasis** *2021 – 2024*
  Zeolite & Nanozeolites In Situ Growth & Nanosilica
  - Engineered **hybrid aerogels** with nanocellulose and inorganic components for enhanced hemostatic coagulation.
  - Tested performance in **femoral bleeding** and **liver laceration** models.
  - Showed superior **clotting efficacy** and **biosafety** profile compared to commercial products.
- **Biomimetic Biomass-Based Hemostatic Materials** *2023 – 2025*
  Advanced Hemostats & Hemostatic Mechanism Investigation
  - Developed biomass-derived materials that **mimic the hemostatic mechanism** of zeolites.
  - Engineered **porous structures** to enhance fluid absorption and promote concentration of clotting factors.
  - Evaluated **biocompatibility and hemostatic efficiency** in *in vitro* and *in vivo* models, demonstrating effective bleeding control.

SKILLS

- **Laboratory Techniques:** SEM, TEM, Dynamic Light Scattering (DLS), Rheometry, X-ray Diffraction (XRD), Universal Testing/DMA, Laser Confocal Scanning Microscopy (CLSM), Cell Culture, Animal Bleeding Model
- **Software:** Adobe Illustrator, Origin, GraphPad Prism, ImageJ, Blender
- **Languages:** Japanese (CEFR A1), English (CET-4: 614, CET-6: 576), Mandarin Chinese (Native)
- **Certifications:** High School Chemistry Teaching Certificate, CATTI (Translation) Level 3, LSCAT s220

PUBLICATIONS

1. Jin, Z.; **Chen, L.**; Xia, R.; Gong, Y.; Zhang, H.; Wang, G.; Zhang, Q. *Zeolite-Mimetic Hemostatic Gauze Based on Calcium-Chelated Cellulose Nanofibers for Efficient and Safe Hemostasis*. **Small** 2026, e74902. https://doi.org/10.1002/smll.74902.
2. **Chen, L.**; Xia, R.; Jin, Z.; Li, W.; Wang, G.; Zhang, Q. *Calcium-Releasing Silica-Coated Cellulose Nanofiber Aerogel for Safe and Efficient Hemostasis*. **Biomater. Adv.** 2026, 215061. https://doi.org/10.1016/j.bioadv.2026.215061.
3. Wang, J.; Chang, Y.; Liu, X.; **Chen, L.**; Tian, Y.; Huang, X.; Zhang, Q. *A Highly Ductile and Multifunctional Cellulose/$CaCl_2$ Composite Film for Humidity-Responsive Wearable Sensors*. **Int. J. Biol. Macromol.** 2026, 358, 151743. https://doi.org/10.1016/j.ijbiomac.2026.151743.
4. Liu, X.; Tian, Y.; Kang, S.; Jin, Z.; **Chen, L.**; Lu, Q.; Wang, L.; Zhang, Q. *Mechanically Robust, Forgeable, and Machinable Cellulose Hydrogel*. **ACS Sustainable Chem. Eng.** 2026. https://doi.org/10.1021/acssuschemeng.5c13794.
5. Xia, R.; Wang, L.; **Chen, L.**; Liu, X.; Chang, Y.; Tian, Y.; Jin, Z.; Zhang, Q. *Facile Synthesis of Regenerated Chitosan Nanowhiskers with Concentration-Dependent Self-Assembly Behavior*. **Carbohydr. Polym.** 2026, 124615. https://doi.org/10.1016/j.carbpol.2025.124615.
6. Chang, Y.; Tian, Y.; Wang, J.; Zhao, J.; **Chen, L.**; Kang, S.; Lu, Q.; He, X.; Zhang, Q. *High-Elongation, Water-Weldable, and Fully Degradable Biomass Foams Fabricated via Oven Drying*. **Sci. Adv.** 2025, 11 (31), eady0746. https://doi.org/10.1126/sciadv.ady0746.
7. **Chen, L.**; Jin, Z.; Kamiya, T.; Liu, X.; Tian, Y.; Xia, R.; Wang, G.; Li, T.; Zhang, Q. *Nanoporous Zeolite Anchored Cellulose Nanofiber Aerogel for Safe and Efficient Hemostasis*. **Small** 2025, 21 (21), 2500696. https://doi.org/10.1002/smll.202500696.
8. Tian, Y.; **Chen, L.**; Liu, X.; Chang, Y.; Xia, R.; Zhang, J.; Kong, Y.; Gong, Y.; Li, T.; Wang, G.; Zhang, Q. *Colored Cellulose Nanoparticles with High Stability and Easily Modified Surface for Accurate and Sensitive Multiplex Lateral Flow Assay*. **ACS Nano** 2025, 19 (4), 4704–4717. https://doi.org/10.1021/acsnano.4c15340.
9. Wang, L.; Tian, Y.; Chang, Y.; **Chen, L.**; Zhang, Q. *Cellulose Nanofiber-Created Air Barrier Enabling Closed-Cell Foams Prepared via Oven-Drying*. **Carbohydr. Polym.** 2025, 351, 123096. https://doi.org/10.1016/j.carbpol.2024.123096.
10. **Chen, L.**; Pan, S.; Wang, L.; Liu, X.; Jin, Z.; Xia, R.; Chang, Y.; Tian, Y.; Gong, Y.; Wang, G.; Zhang, Q. *Superhydrophobic Cellulose Nanofiber Aerogels for Efficient Hemostasis with Minimal Blood Loss*. **ACS Appl. Mater. Interfaces** 2024, 16 (36), 47294–47302. https://doi.org/10.1021/acsami.4c10505.
11. Jin, Z.; **Chen, L.**; Liu, X.; Xia, R.; Li, W.; Wang, G.; Zhang, Q. *Zeolite Firmly Anchored Regenerated Cellulose Aerogel for Efficient and Biosafe Hemostasis*. **Int. J. Biol. Macromol.** 2025, 304, 140743. https://doi.org/10.1016/j.ijbiomac.2025.140743.
12. Feng, Y.; Chang, Y.; Wang, L.; Liu, X.; **Chen, L.**; Yan, X.; Zhang, Q. *Ultrahigh-Strength Cellulose Nanofiber Foams via the Synergy of Freeze-Casting and Solvent Exchange*. **Carbohydr. Polym.** 2025, 347, 122671. https://doi.org/10.1016/j.carbpol.2024.122671.
13. Tian, Y.; Kong, Y.; Liu, X.; **Chen, L.**; Wang, L.; Zhou, L.; Wang, G.; Zhang, Q. *Regenerated Carboxymethyl Cellulose Beads for Selective Removal of Low-Density Lipoprotein from Whole Blood*. **Carbohydr. Polym.** 2025, 348, 122848. https://doi.org/10.1016/j.carbpol.2024.122848.
14. Liu, X.; Tian, Y.; Wang, L.; **Chen, L.**; Jin, Z.; Zhang, Q. *A Cost-Effective and Chemical-Recycling Approach for Facile Preparation of Regenerated Cellulose Materials*. **Nano Lett.** 2024, 24 (29), 9074–9081. https://doi.org/10.1021/acs.nanolett.4c02351.
15. Pan, S.; Li, Y.; Tong, X.; **Chen, L.**; Wang, L.; Li, T.; Zhang, Q. *Strongly-Adhesive Easily-Detachable Carboxymethyl Cellulose Aerogel for Noncompressible Hemorrhage Control*. **Carbohydr. Polym.** 2023, 301, 120324. https://doi.org/10.1016/j.carbpol.2022.120324.

PATENTS

1. Zhang, Q.; **Chen, L.**; Jin, Z.; Tian, Y. *Preparation Method of In Situ Grown Nano-Zeolite Nanocellulose Aerogel and Its Hemostatic Application*. **Patent** 2025, Patent No. 202510141274.0.
2. Zhang, Q.; Jin, Z.; **Chen, L.** *Preparation of Calcium-Ion-Bound Carboxymethyl Nanocellulose Gauze and Its Hemostatic Application*. **Patent** 2025, Patent No. 202510497801.1.
3. Zhang, Q.; Tian, Y.; Li, T.; **Chen, L.** *Colored Cellulose Nanoparticles and Their Preparation Method and Application*. **Patent** 2024, Patent No. 202410928182.2.
4. Zhang, Q.; Tian, Y.; Wang, L.; Chang, Y.; **Chen, L.** *Cellulose Foam Materials and Their Preparation Method and Application*. **Patent** 2024, Patent No. 202410928190.7.

Leadership Experience

- **Lab Manager** *2021 – 2024*
  School of Life Sciences, East China Normal University
  - Managed lab inventory, procurement, and vendor negotiation.
  - Supervised and trained undergraduate assistants on safety protocols, equipment operation, and data workflows.
  - Implemented a digital scheduling system for instrument bookings, increasing equipment utilization efficiency.
  - Coordinated cross-laboratory collaborations.
- **Chair, Graduate Student Union** *2020 – 2021*
  School of Life Sciences, East China Normal University
  - Oversaw academic seminars, community outreach, and social events for over 200 graduate students.
  - Enhanced departmental engagement by introducing new student-led symposiums.
- **Director, New Media Center** *2016 – 2017*
  School of Sciences, Zhejiang A&F University
  - Managed digital platforms and university-wide communications.
  - Organized promotional campaigns for academic conferences and cultural festivals.

Selected Honors and Awards

- **Postdoctoral Fellowship Program (Grade C)** *2026*
  China Postdoctoral Science Foundation
- **Distinguished Graduate Student of Shanghai** *2025*
  Shanghai Municipal Education Commission
  - Recognized as one of the top graduates in Shanghai for outstanding academic and extracurricular achievements.
- **National Scholarship (Ph.D. Level)** *2024*
  Ministry of Education, China
  - Awarded for exceptional academic performance and research excellence during doctoral studies.
- **National First Prize, 19th Challenge Cup Special Contest** *2024*
  Central Committee of the Communist Youth League, China Association for Science and Technology
  - Participated in a team project on advanced nanocellulose-based materials.
- **Outstanding Graduate Student Union Leader** *2020*
  East China Normal University
  - Honored for leadership in organizing academic and cultural events.
- **Outstanding Student** *2019*
  East China Normal University
  - Recognized for exemplary academic performance and campus involvement.
- **Provincial Second Prize, 15th Challenge Cup National College Student Academic Works Competition** *2017*
  Zhejiang Provincial Committee for College Student Science and Technology Competition
  - Awarded for an innovative extracurricular research project.

Professional Memberships

- **Chinese Society for Biomaterials (CSBM),** Member ID: U55OBT7c *2024 – Present*
- **Chinese Chemical Society (CCS),** Member ID: 220511220 *2022 – Present*

References

1. **Prof. Min Zhou**
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   Postdoctoral Supervisor
2. **Prof. Qiang Zhang**
   School of Life Sciences, East China Normal University
   qzhang@bio.ecnu.edu.cn
   Ph.D. Supervisor
3. **Prof. Linfang Shi**
   College of Chemistry and Materials Engineering, Zhejiang A&F University
   lfshi2003@163.com
   Undergraduate Tutor

Additional Information

Date of Birth: August 5, 1997
Current Position: Postdoctoral Researcher, Zhejiang University-University of Edinburgh Institute, International Campus, Zhejiang University